

DZUHAIR HAKIMI

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Passionate about creating engaging and innovative programs where users are encouraged to think creatively to achieve their goals. Graduate Computer Science student looking for a position in Data Analysis, Quality Assurance or Software Engineering.

EDUCATION

Bachelor of Science (Hons) Computing Science

Sep 2023 – May 2026

Heriot-Watt University, Edinburgh, Scotland

- Current Grade: **First Class**
- Awarded Deputy Principal's Award
- Relevant Courses: Database Management Systems, Software Engineering, Data Visualization and Analytics, Computer Network Security, Artificial Intelligence and Intelligent Agents, Video Games Programming, Honours Project.

Foundation in Science

Heriot-Watt University Malaysia, W.P. Putrajaya

Apr 2022 – Apr 2023

- GPA: 3.5

EXPERIENCE

Software Testing Analytics Intern

May 2025 – Aug 2025

Millennium Radius Sdn Bhd

Kuala Lumpur, Malaysia

- Planned, executed, and documented internal tests to validate functional and non-functional requirements across diverse software applications.
- Uphold quality assurance by creating detailed test cases and managing data entry within Excel sheets.
- Collaborated directly with developers to ensure the correctness of data and error-free content implementation.
- Assisted senior QA testers in performing User Acceptance Tests, maintaining a rigorous approach to accuracy.
- Authored detailed test summary reports and documentation for review and presentation.

SKILLS

Spreadsheet and Analysis: MS Excel, Google Sheets, Tableau, PowerBI, SQL

Programming & Web Development: Java, C++, C#, Python, JavaScript, HTML, CSS

Game Engines: Unity, Unreal Engine, Godot, GameMaker Studio

Other Tools: MS 365, Notion, Figma, Canva, Blender, WebGPU

EXTRACURRICULAR

Treasurer – Heriot-Watt University Ravagers Football Club

Sep 2024 – May 2025

- Kept accounts on the club's incomes and expenses, planned for events for fundraising, and book fields for friendly sessions and team training.

Event Organizer – Footie for Fugees

Sep 2023 – Mar 2024

- Organized events with a team of 10 like-minded people to raise funds for the Fugees School including a friendly football session with the school, multiple booths and a charity match.

PROJECTS

Real-time Destruction in Video Games

- Engineered a real-time mesh destruction system in **Unity 3D** utilizing Nvidia PhysX, Voronoi diagrams, and Constructive Solid Geometry (CSG) to simulate realistic, physics-based fractures.
- Benchmarked system performance and scalability against industry-standard, pre-fractured modeling techniques.

Interactive Data Story: East Asia & Pacific

- Co-developed dynamic data visualizations using **D3.js** and **Vanilla JS** to explore World Bank demographic trends across 36 countries building an **interactive website** featuring connected scatter plots, choropleth maps, and stacked area charts with dynamic filtering and tooltips.
- Executed EDA on initial data including **data cleaning, visualizing data with Tableau, and presenting insights and findings.**

Smart Home System

- Collaborated in an 8-person team to build a **smart home web application**, leading frontend development and integrating ChartJS to visualize real-time energy data.
- Acted as a liaison between group members and manager to show progress, track milestone and find issues as well as demonstrated the software in front of an audience.

Sudoku Solver

- Co-developed a Sudoku solver in **Python** with a custom GUI, utilizing **backtracking and pruning algorithms** to outperform standard brute-force and heuristic methods.

PDDL Planner

- Co-created an automated lunar rover planner using **PDDL**, successfully defining domain objects and actions to simulate various mission parameters and end states.

World List Map

- Developed a file parsing program utilizing custom **Data Structures and Algorithms** (linked lists) to efficiently map, search, and track word frequencies across multiple files.

Public-Key Encryption Using Merkle-Hellman Knapsack Cryptosystem

- Developed a **Python-based public-key cryptography program**, implementing custom algorithms for key-pair generation, message encryption, and decryption utilizing super increasing integer sequences.
- Engineered a **secure decryption mechanism** using modular arithmetic and mathematical functions (such as greatest common divisor and modular inverses) to efficiently decode n-bit ciphertexts back into original plaintexts.

Simple Firewall Manager Implementation

- Engineered a **Python-based command-line firewall manager** to dynamically create, delete, and query IPv4 access control lists based on packet direction and strict priority ordering.
- Implemented **priority-based network filtering logic** to evaluate incoming and outgoing traffic, supporting both exact IP address matching and localized address range evaluations.

Battle Builders

- Designed the end-to-end game flow for a physics-based combat/building game, utilizing paper **prototype and playtesting** feedback to iteratively balance combat loops.
- Implemented a digital prototype using **Unity 2D** specifically the build system where players choose which block to place and decide where to place it. Also implemented controls for local multiplayer and basic UI.

Animal Farm

- Developed a Unity 3D game utilizing rigidbody physics and **AI NavMesh**, implementing complex, multi-state animal behaviors and flocking algorithms (boids).

Obstacle Assault

- Programmed a 3D platformer in **Unreal Engine** utilizing **C++ and Blueprints** to build custom, reusable classes for moving and rotating platforms.

Jump 'em JON

- Authored a comprehensive **Game Design Document** detailing UI flows, environmental traps, and a balanced 5-stage progression system designed to teach complex mechanics.

Warehouse Wreckage

- Created an **Unreal Engine** physics-based arena using Blueprints, implementing object collision, projectile spawning, and ammo management.

Squash the Creep 3D

- Built a 3D platforming game in **Godot using C#**, successfully translating 2D movement logic into a 3D space with scoring and reset mechanics.

Dodge the Creep 2D

- Developed a 2D evasion game in **Godot using C#**, implementing core mechanics like player movement, collision detection, and a complete UI loop.

Infinite Brick Breaker

- Developed a Brick Breaker game in **GameMaker Studio** where the player has to last as long as possible, implementing movement, UI, scores, and powerups.

3D Train Station Diorama

- Collaborated in a 3-person team to create a diorama of a three-tiered train station working on the bottom level platform and train.
- Diorama is created as a **Blender** model and **Unreal Engine** animation as well as an **interactive web application** by using JavaScript for the website and WebGL as a renderer.